



Effects of oral herbal powder supplements on the adaptive capabilities of heat stressed Kenguri sheep based on changes in physiological and molecular response

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Abstract

A study was designed to establish the effect of heat stress on the adaptive capability of Kenguri sheep and to test the efficacy of herbal powder supplement for relieving heat stress in this breed. The study was carried out for a period of sixty days in twenty-four one-year-old Kenguri ewes which were randomly distributed into 4 different groups with six animals each as follows; KC ($n=6$; Kenguri Control), KHS ($n=6$; Kenguri Heat Stress), KCS ($n=6$; Kenguri Control herbal Supplement) and KHSS ($n=6$; Kenguri Heat Stress herbal Supplement). The results on temperature-humidity index (THI) indicated that comfort condition was provided for KC and KCS while severe heat stress condition was experienced by KHS and KHSS groups. Heat stress significantly increased the respiratory rate (RR), pulse rate (PR) and rectal temperature (RT). Further, heat stress also significantly upregulated the Heat Shock Transcription Factor 1 (*HSF1*), Heat Shock Protein 27 (*HSP27*), *HSP70*, *HSP90* and *HSP110*. These results indicate that RR, PR, RT, *HSF1*, *HSP27*, *HSP70*, *HSP90*, and *HSP110* can serve as reliable indicators for quantifying heat stress response in Kenguri sheep. In addition, the results also established the heat stress-relieving effects of herbal supplements as reflected by the significantly reduced physiological responses recorded and significantly downregulated all the genetic traits studied. Thus, it could be concluded that the herbal supplements used in the study was found to have effective potential to reduce the adverse impacts of heat stress in Kenguri sheep.

Keywords Adaptation · Heat stress · Herbal product · Hsps · Sheep · THI

Introduction

The global climate change phenomenon has been well established to be a serious long-enduring challenge for livestock producers, impacting livestock health and production (Silanikove and Koluman 2015). Given the importance of livestock in supporting the rural economy and in ensuring food security, in this climate change scenario it is crucial to ensure optimal livestock production that supports the predicted rise in human population by 2050 (United Nations Population Division 2019). To face the upcoming challenges associated with the demanding climate change scenario, small ruminant production has been found suitable as they are well-adapted to harsh environments (Silanikove 2000; Leite et al. 2021). This study in Kenguri sheep is the first of its kind in establishing the adaptive capacity of the particular breed to heat stress challenges. The study's

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hypothesis was justified by studying the changes associated with their physiological responses and expression profile of selected molecular chaperones. The information generated from this study could be very valuable to establish the climate resilience of this indigenous breed.

Sheep have been well established to be bestowed with unique morphological, behavioural, physiological and genetic adaptive traits to combat heat stress challenges (Berihulay et al. 2019). The physiological adjustments like changes in rectal temperature, pulse rate, skin temperature and respiration rate have been used as reliable indices to assess heat stress in sheep (Al-Haidary 2004; Indu and Pareek 2015; Al-Dawood 2017). In order to tackle the negative impacts of heat stress, sheep demonstrate adaptive response at the molecular level directed by altered expression patterns of heat shock proteins (HSPs) that function as cellular chaperones, preventing protein and cellular damage during heat exposure (Joy et al. 2020). Further, according to Romero et al. (2013), sheep subjected to greater ambient temperatures have been shown to produce more HSPs as an adaptive strategy to prevent damage to proteins and cells. The thermo-tolerant capability of sheep breeds can be identified using these molecular markers, aiding in re-strategizing breeding programs. Indigenous sheep breeds native to arid and semi-arid conditions are demonstrated to perform better in harsh environmental conditions than non-native breeds (Chedid et al. 2014). The indigenous breeds with an innate capability to adapt to climate change can be screened as they act as significant resources for breeding characteristics related to climate resilience (Joy et al. 2020; Wanjala 2024). The key strategy of identifying the differences in adaptation between indigenous breeds and exotic breeds can provide significant clues to develop solutions for problems caused by climate change in sheep production systems.

The interaction of the sheep husbandry system with environmental variables in line with the climate change scenario prompts urgent mitigation policies. According to Naqvi et al. (2017), the major ways to mitigate adverse heat stress effects in animals maintained in hot environment are housing management, enhancement of nutritional management, and genetic production of heat-resistant breeds. Among the various mitigation approaches, nutritional interventions have been proven to be effective in relieving heat stress in sheep (Sejian et al. 2024). The herbal mixtures have anti-inflammatory, antioxidant, and anti-microbial effects, posing them to be a potential nutritional intervention and an ideal nutritional supplement for heat stressed animals (Bhatt 2015). Further, the complicated pathogenesis of heat stress in animals often means, a single herbal remedy proves insufficient in eliciting a significant impact, instead, polyherbal preparations may exert a synergistic action, improving the overall

effect on heat-stressed cells (Zhao et al. 2022). However, there is a paucity of information related to the use of herbal supplement to mitigate heat stress in sheep. The hypothesis of the present study is to unravel the potential use of herbal supplements in improving the adaptive capabilities of sheep under heat load conditions. Hence, the objective of the study was to investigate the benefits of the herbal supplement in modulating the adaptive response of heat-stressed indigenous Kenguri sheep. Additionally, this study attempts to establish the heat tolerant capability of the locally adapted Kenguri breed based on the adaptive responses.

Materials and methods

Experimental location and climate

The study was conducted at the Experimental Livestock Unit (ELU), ICAR-National Institute of Animal Nutrition and Physiology (Bangalore, India), which is located in the southern Deccan plateau of the country, at longitude 12°57'04.3"N, latitude 77°36'25.3"E and positioned at an elevation of 920 m above mean sea level (MSL). The average annual ambient temperature fluctuates between 15 and 36 °C. The average annual relative humidity (RH) ranges from 20 to 85%. The area receives 200 to 970 mm of rain annually and is distributed erratically throughout the year. The experimental trial was carried out between June and August. The annual minimum and maximum temperatures vary between 15 and 22 °C and 27–34 °C respectively. The annual RH fluctuated between 28 and 59 per cent throughout the research period amid hot semi-arid conditions.

Animals

Kenguri, also known as Tenguri, is a large-sized sheep breed that prevails in the Northern dry Zone of Karnataka, with a tropical climate. This breed is primarily reared for mutton, in addition to income generation through the sale of milk and dung. The breed has a characteristic dark brown-coloured coat with droopy ears and an average adult body weight of 32–72 kg (NBAGR 2020). The present study was conducted with 24 adult (one year old) female Kenguri sheep weighing 15–23 kg (P value of 0.732). The animals were maintained in well-ventilated, hygienic barns with asbestos ceilings. The animals were supplied with good-quality feed and drinking water. To ensure animal health throughout the study period, prophylactic procedures (Vaccination and deworming) against sheep diseases such as sheep pox, peste des petits ruminants, enterotoxemia, endo- and ectoparasitic infestations were carried out according to the standard health management schedule (Srikanth et al. 2023).

Table 1 Feed ingredients and chemical composition of concentrate and roughage fed to the experimental animals

Attribute	Concentrate mixture (kg/100 kg)	Napier hay (Pennisetum purpureum)
Ingredients		
Maize	36	-
Wheat bran	37	-
Soybean meal	25	-
Mineral mixture	1.5	-
Salt	0.5	-
Chemical composition (%)		
Dry matter	90.83	92.32
Organic matter	94.02	88.59
Crude protein	23.15	6.84
Ether extract	3.77	2.72
Crude Fibre	6.12	32.58
Total ash	5.98	11.41
Fibre fractions (%)		
Neutral detergent fibre	30.94	69.55
Acid detergent fibre	8.92	42.67
Nutritive value		
Total digestible nutrients (%)	71	53.20
Digestible energy (kJ/kg)	14.24	9.78
Metabolizable energy (kJ/kg)	11.78	7.98

Experimental design

The study was carried out for 60 days in a controlled climate chamber housing condition. The experimental animals were randomly assigned into four groups with six animals each, viz., KC ($n=6$; Kenguri Control; standard diet), KHS ($n=6$; Kenguri Heat Stress; standard diet), KCS ($n=6$; Kenguri Control Supplement; standard diet+herbal powder supplement), and KHSS ($n=6$; Kenguri Heat Stress Supplement; standard diet+herbal powder supplement). The animals were completely stall-fed with a diet composed of 60% roughage (Hybrid Napier) and 40% concentrate (maize, 36 kg/100kg; wheat bran, 37 kg/100kg; soybean meal, 25 kg/100kg; mineral mixture, 1.5 kg/100kg and common salt, 0.5 kg/100kg). The commercial mineral mixture having the composition of calcium, phosphorous, manganese sulphate, potassium iodide, ferrous sulphate, copper sulphate, cobalt sulphate, zinc sulphate, and selenium was used. Table 1 details the feed ingredients and chemical composition of concentrate and roughage fed to the experimental animals.

Climate chamber and animal housing details

The climate chamber, spanned 12.8 m in length, 3.41 m in width, and 2.87 m in height (total area: 143.00 m²) and was constructed with anti-corrosive laminated steel sheets and insulated with polystyrene foam. The facility can accommodate up to 12 animals, each housed in individual pens measuring 1.95 m² each. These pens are elevated 0.6 m above the main floor to facilitate hygiene and waste management. Further, these pens are separated by stainless steel rods to prevent physical movement between pens, while still allowing visual contact to minimize socialization stress. Each pen includes a separate feeder, waterer, and trays for collecting urine and faeces, and animals were kept in individual pens throughout the study.

Two types of chambers that provides appropriate environmental condition for any environmental stress-related studies was used: one simulating a thermal comfort zone and the other designed to induce heat stress. The thermoneutral zone (TNZ) chamber was maintained at 24 °C, within the established optimal temperature range for thermal comfort in sheep (20–28 °C) (de Sousa Silva et al. 2024). The TNZ chamber was maintained at 24 °C, for six hours daily (10:00 h to 16:00 h), providing ideal conditions for the KC and KCS group ewes. The heat stress chamber exposed KHS and KHSS group ewes to a simulated heat stress model tabulated in Table 2 and described by Silpa et al. (2023). The model was developed using 30 years of historical temperature and humidity data for the region's summer months (April to June). The 6-hour exposure period was selected to mirror natural heat stress experienced during grazing in extensive systems. From an animal welfare standpoint, this approach was more ethically sound and less distressing than constant heat exposure models. Figure 1 describes the thermoneutral and heating climate chambers while Fig. 2 describes the housing of Kenguri sheep in individual pen in climate chamber.

The herbal powder mixture composed of *Ocimum sanctum* (Tulsi), *Embllica officinalis* (Amla), *Morinda citrifolia* (Noni), *Withania somnifera* (Ashwagandha) and *Phyllostachys edulis* (Bamboo), was supplemented to KCS and KHSS group animals. The inclusion criteria for these herbs were based on their reported adaptogenic, analgesic, immunomodulatory, hepatoprotective, cardioprotective and antioxidant properties in heat stressed livestock (Lakhani et al. 2016; Su et al. 2024; Verma et al. 2024). The Table 3

Table 2 The simulated heat stress model used for evaluating the adaptive capabilities of heat stressed and herbal powder supplemented Kenguri ewes

Time (h)	10.00	11.00	12.00	13.00	14.00	15.00	16.00
Temperature (°C)	36	37.5	38.5	38.8	40	38.3	37
Relative Humidity (%)	49	44	41	40	39	42	45



Fig. 1 Photograph depicting the thermoneutral and heating climate chambers

describes the composition of the herbal powder supplements. In the current study, the supplement was provided in dry powder form at a dose rate of 0.8 g/Kg BW/Day along with the concentrate ration. The composition, proportion and dosage were calculated based on a comprehensive review of existing literature. Further, all the five herbal ingredients were procured in powdered form and mixed uniformly according to the proportions detailed in Table 3, to prepare 100 g formulation of the herbal mixture. The final dose rate of the herbal supplement was 0.8 g/kg body weight per day which contained the same proportions of the extracts outlined in Table 3. Furthermore, the bioactive compounds of the herbal ingredients were not quantified in the current study.

The animals were maintained in their respective chambers for six hours a day between 10:00 h to 16:00 h in the climate chamber during the experimental period. The experimental animals were fed and watered individually in their respective chambers. All the cardinal weather variables were recorded twice daily (8:00 h and 14:00 h) by following

standard procedures. The THI was calculated based on weather variables using McDowell (1972) formula. The physiological response recording and blood sampling were carried out at fortnightly intervals. Blood collection was carried out on the last day of the experiment and subjected to PBMC isolation for gene expression studies.

Blood collection and PBMCs isolation

The blood samples (5 ml) were collected on the last day of the experiment from the external jugular vein in EDTA vacutainers and stored in ice. Immediately the samples were processed for Peripheral Blood Mononuclear Cells (PBMCs) isolation. To the samples ice-cold $1\times$ RBC lysis buffer was added and incubated at room temperature for 10 min and centrifuged at 2,500 rpm at 4 °C for 5 min. The GeneJET Whole Blood RNA Purification Mini Kit (Thermo Scientific, Lithuania) was used to isolate total RNA from the PMBCs following the manufacturer's protocol with minor modifications as follows: the pellet was re-suspended in 600



Fig. 2 Photograph depicting the individual pen where the Kenguri sheep were housed during the experiment period

Table 3 Composition of the herbal supplement used to relieve heat stress in Kenguri ewes

Nutrient	Composition (100 g)
<i>Ocimum sanctum</i>	30 g
<i>Emblca officinalis</i>	30 g
<i>Morinda citrifolia</i>	20 g
<i>Withania somnifera</i>	10 g
<i>Phyllostachys edulis</i>	10 g
Dose rate	0.8 g/Kg BW/Day

μL of lysis buffer containing 20 μL of 2-mercaptoethanol, mixed well by vortexing. Further, 600 μL of buffer containing 590 μL of Tris EDTA and 10 μL proteinase K was added, vortexed and incubated for one hour. Four hundred and fifty microliters of ethanol (96–100%) were added and mixed by vortexing. The lysate was added to the purification column and centrifuged at 12,000 rpm for 2 min, and the flow-through was discarded. The column and the collection tube were reassembled, and the remaining lysate was transferred, centrifuged as before, and the collection tube containing the flow-through was discarded. Then, 700 μL

of wash buffer 1 was added to the purification column and centrifuged for 2 min at 12,000 rpm, and the flow through was discarded. Six hundred microliters of wash buffer 2 was added and centrifuged at 12,000 rpm for 2 min, and the flow through was discarded. 500 μL of wash buffer 2 was added to the purification column and centrifuged, and the flow through was discarded. The empty purification column was centrifuged again at 12,000 rpm for 2 min. The nuclease-free water (NFW) of 20 μL was added at the midpoint of the column and centrifuged for 5 min at the rate of 12,000 rpm. The RNA eluted was collected in fresh nuclease-free microcentrifuge tubes and processed immediately.

The total RNA was reverse transcribed into complementary DNA (cDNA) using RevertAid First Strand cDNA Synthesis Kit for real-time quantitative polymerase chain reaction (RT-qPCR) (Thermo Scientific, Lithuania). This was followed as per the manufacturer's protocol as follows: 4 μL of 5 $\times$ Reaction Mix, 1 μL RevertAid M-MuLV Reverse Transcriptase (RT) (200 U/ μL), 5 μg of template RNA, 15–20 pmol of primer, 1 μL of RiboLock RNase

Inhibitor (20 U/ μ L), 2 μ L of 10 mM dNTP Mix and NFW was added to make the volume to 20 μ L. Then, the contents were mixed gently and subjected to reverse transcribing PCR (5 min at 25 °C, followed by 42 °C for 1 h, and the enzymes were inactivated by heating at 70 °C for 5 min). The product of the first-strand cDNA synthesis was diluted to a final concentration of 25 ng/ μ L with NFW and 2 μ L of diluted cDNA was used for each reaction in qPCR. Gene-specific primer was designed using online NCBI primer design software and specificity was checked using Primer3 and BLAST (<http://www.ncbi.nlm.nih.gov/tools/primer-blast/>). Table 4 describes the primer sequences of the housekeeping gene and target genes.

The relative expression of selected genes was studied using SYBR green chemistry (Maxima SYBR green qPCR master mix, Fermentas, USA). The 20- μ L reaction was carried out in triplicate using 50 ng of template and 0.5 μ M primer concentrations. The reaction condition was enzyme activation at 95 °C for 10 min and amplification cycle (40 cycles; initial denaturation at 95 °C for 15 s, annealing at 60 °C for 30 s, and extension at 72 °C for 30 s). At the end of the reaction the non-specific amplification was ruled out by checking the melt curve analysis. The GAPDH (glyceraldehyde 3-phosphate dehydrogenase) housekeeping gene was used as internal control to compare with the relative expression of target genes. The relative expression patterns of target genes were analyzed as per the $2^{-\Delta\Delta CT}$ formula.

Variables studied

The THI was calculated as per the formula outlined by McDowell (1972). The physiological responses such as respiration rate (RR), pulse rate (PR) and rectal temperature (RT) were recorded at 8:00 h and 14:00 h, at fortnightly intervals. The RR was recorded based on observing the flank movements for one minute using a stopwatch, approximately 4–5 m away from the animal. The unit of measurement of RR was in breaths/min. The PR was recorded by palpation of the femoral artery for one minute. The unit of

measurement of PR was in beats/min. The RT was recorded using a digital rectal thermometer. The unit of measurement of RT was in °C. The targeted gene expression patterns were Heat Shock Transcription Factor 1 (*HSF1*), Heat Shock Protein 27 (*HSP27*), *HSP70*, *HSP90* and *HSP110* genes.

Ethics declaration

All the experimental procedures, including subjecting the animals to heat stress involved in the study, were conducted with the approval from the institute animal ethics committee (Reference number V-11011(13)/14/2021-CPCSEA-DADF).

Data analysis

The physiological variables in the study were subjected to 2×2 factorial analysis with temperature, diet and experimental days as fixed factors. The changes in relative expression of PBMC target genes in relation to GAPDH as the housekeeping gene were analyzed by 2×2 factorial analysis with temperature and diet as fixed factors with Tukey's post hoc analysis to compare the means between the groups. The significance level was set at $P < 0.05$. The data obtained were statistically analyzed using SPSS software (Version 29.0).

Results

Temperature humidity index (THI)

Figure 3 describes the average temperature humidity index during both the morning and afternoon for the study duration in the thermoneutral and heating climate chambers. The THI obtained during 8.00 and 14.00 h for the 60-day duration clearly indicated the significantly ($p < 0.01$) higher THI obtained during 14.00 h in the heating chamber as compared to the thermoneutral chamber. As per McDowell (1972), any THI value above 78 is considered extremely severe heat stress, and any value below 72 is considered comfortable.

Table 4 Primers used for studying the relative expression of the targeted genes in heat stressed Kenguri ewes

Gene ID		Primer Sequence (5'-3')	Product size (bp)	Accession no.
GAPDH	F	GGTGATGCTGGTGCTGAGTAG	147	XM_005680968.3
	R	ACGATGTCCACTTTGCCAGT		
HSF-1	F	AAAGTGACCAGCGTGTCCA	115	NM_001314344.2
	R	AGTCCATGCTCTCCTGCTTC		
HSP27	F	CCTGGACGTCAACCACTTC	76	JQ957566.1
	R	GCTTGCCAGTGATCTCCAC		
HSP 70	F	TGGCTTTCACCGATACCGAG	167	NM_001285703.1
	R	GTCGTTGATCACGCGAAAG		
HSP 90	F	TACCCACGGCATTTCACCA	107	XM_005694067.1
	R	CTCATTAGCCTCGGCATCTGG		
HSP 110	F	AAAGGAGCTCGTCTTGGGAC	142	XM_018056621.1
	R	TACCCACGGCATTTCACCA		

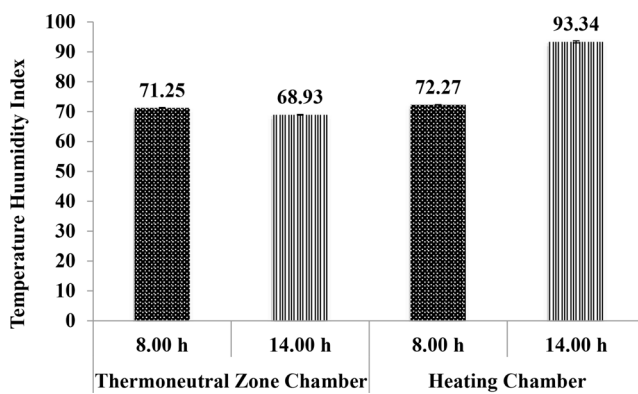


Fig. 3 The average THI both during morning and afternoon in the climatic chambers

The average 93.34 obtained during 14.00 h duration demonstrated the extremely severe heat stress exposure to both KHS and KHSS groups. Further, the 69 THI obtained in the TNZ chamber indicated the comfortable condition for the KC and KCS groups.

Effect of heat stress and herbal powder supplementation in physiological responses in Kenguri sheep

Respiration rate (RR)

Respiration rate was lower in the morning than in the afternoon (26.1 vs. 85.4 breaths/minute, $p < 0.001$). Respiration rate in the morning was different between days, being lowest on day 0 and highest on day 60 (22.6, 27.3, 26.5, 26.7 and 27.2 breaths/minute for days 0, 15, 30, 45, and 60, respectively, $P < 0.001$) (Table 5). There was no significant effect of temperature on respiration rate in the morning (25.6 vs. 25.5 breaths per minute for thermoneutral and heat stress conditions, respectively, $P = 0.08$). Further, there was no significant interaction between temperature and day of treatment. The respiration rate in the afternoon was increased during heat stress (60.0 vs. 110.8 breaths per minute for thermoneutral and heat stress conditions, $P = 0.00$) and was

reduced by the herbal supplement (89.2 vs. 81.6 breaths/min for control and herbal diet, $P < 0.001$) (Table 5). Respiration rate in the afternoon was different between days, being lowest on day 0 and highest on day 60 (25.8, 94.2, 96.0, 107.3 and 103.7 breaths/minute for days 0, 15, 30, 45, and 60, respectively, $P = 0.00$). Further, the diet also significantly influenced the respiration rate in afternoon (89.2 vs. 81.6 breaths/min for control and herbal diet, $P < 0.001$). In addition, the temperature $\times$ diet, temperature $\times$ day, diet $\times$ day and temperature $\times$ diet $\times$ day also significantly ($P < 0.001$) influenced the respiration rate, such that the difference in respiration rate between the thermoneutral and heat stress conditions was greater on day 15, 30, 45 and day 60 than on day 0, while the herbal diet reduced the respiration rate in all those days of recording (Table 5).

Pulse rate (PR)

Pulse rate was lower in the morning than in the afternoon (68.4 vs. 87.1 beats/minute, $p < 0.001$). Pulse rate in the morning was different between days, being lowest on day 0 and highest on day 45 (67.8, 68.3, 68.7, 69.0 and 68.3 beats/minute for days 0, 15, 30, 45, and 60, respectively, $P < 0.001$) (Table 6). There was no significant effect of temperature on pulse rate in the morning (67.9 vs. 69.0 beats/minute for thermoneutral and heat stress conditions, respectively, $P = 0.154$). Further, both diet (69.3 vs. 67.6 beats/minute for control and herbal diet, respectively, $P = 0.023$) and interaction between temperature and day of treatment ($P < 0.001$) were significant. The pulse rate in the afternoon was increased during heat stress (77.8 vs. 96.5 beats per minute for thermoneutral and heat stress conditions, $P = 0.00$) and was reduced by the herbal supplement (90.3 vs. 84.1 beats /min for control and herbal diet, $P = 0.004$) (Table 6). Pulse rate in the afternoon was different between days, being lowest on day 60 and highest on day 0 (101.4, 96.2, 95.9, 73.2 and 69.2 beats /minute for days 0, 15, 30, 45, and 60, respectively, $P < 0.001$). Further, the diet also significantly influenced the pulse rate in the afternoon (90.3 vs. 84.1 beats /min for control and herbal diet, $P = 0.004$).

Table 5 Effect of heat stress and herbal products supplementation on the respiration rate in Kenguri ewes

Time	Temperature (T)	Diet	Day					Significance
			0	15	30	45	60	
AM	Thermoneutral	Control	21.8 $\pm$ 1.12	26.0 $\pm$ 1.35	27.3 $\pm$ 1.33	26.7 $\pm$ 0.92	26.7 $\pm$ 1.13	Day***
		Herb	24.0 $\pm$ 1.12	26.7 $\pm$ 1.35	24.0 $\pm$ 1.33	26.0 $\pm$ 0.92	26.7 $\pm$ 1.13	
	Heat Stress	Control	22.7 $\pm$ 1.12	28.7 $\pm$ 1.35	28.7 $\pm$ 1.33	28.0 $\pm$ 0.92	29.3 $\pm$ 1.13	T***, Diet***, Day***, T.Diet***, T.Day***, Diet.day***, T.Diet.Day***
		Herb	21.8 $\pm$ 1.12	28.0 $\pm$ 1.35	26.0 $\pm$ 1.33	26.0 $\pm$ 0.92	26.0 $\pm$ 1.13	
PM	Thermoneutral	Control	24.2 $\pm$ 0.36	58.7 $\pm$ 2.82	46.7 $\pm$ 1.75	90.7 $\pm$ 1.93	87.3 $\pm$ 1.75	T***, Diet***, Day***, T.Diet***, T.Day***, Diet.day***, T.Diet.Day***
		Herb	26.7 $\pm$ 0.36	52.7 $\pm$ 2.82	44.7 $\pm$ 1.75	85.3 $\pm$ 1.93	82.7 $\pm$ 1.75	
	Heat Stress	Control	27.0 $\pm$ 0.36	132 $\pm$ 2.82	150 $\pm$ 1.75	137 $\pm$ 1.93	138 $\pm$ 1.75	T***, Diet***, Day***, T.Diet***, T.Day***, Diet.day***, T.Diet.Day***
		Herb	25.5 $\pm$ 0.36	133 $\pm$ 2.82	143 $\pm$ 1.75	116 $\pm$ 1.93	107 $\pm$ 1.75	

*** Significant at $P < 0.001$; ** Significant at $P < 0.01$; * Significant at $P < 0.05$; NS-Non Significant

Table 6 Effect of heat stress and herbal products supplementation on the pulse rate in Kenguri ewes

Time	Temperature (T)	Diet	Day					Significance
			0	15	30	45	60	
AM	Thermoneutral	Control	70.3±1.91	66.3±1.20	69.3±1.90	71.3±1.61	66.7±1.73	Diet*, T.Day***
		Herb	69.3±1.91	63.3±1.20	68.8±1.90	64.7±1.61	68.7±1.73	
	Heat Stress	Control	65.7±1.91	71.3±1.20	68.0±1.90	72.7±1.61	71.3±1.73	
		Herb	66.0±1.91	72.0±1.20	68.7±1.90	67.3±1.61	66.7±1.73	
PM	Thermoneutral	Control	99.3±2.38	89.3±8.17	99.3±3.52	53.3±1.61	51.3±1.42	T***, Diet**, Day***, T.Diet*, T.Day***, Diet.Day***, T.Diet.Day**
		Herb	101±2.38	109±8.17	86.0±3.52	46.0±1.61	43.3±1.42	
	Heat Stress	Control	100±2.38	102±8.17	104±3.52	103±1.61	101±1.42	
		Herb	105±2.38	84.0±8.17	94.7±3.52	90.0±1.61	81.3±1.42	

*** Significant at $P<0.001$; ** Significant at $P<0.01$; * Significant at $P<0.05$; NS-Non Significant

Table 7 Effect of heat stress and herbal products supplementation on the rectal temperature in Kenguri ewes

Time	Temperature (T)	Diet	Day					Significance
			0	15	30	45	60	
AM	Thermoneutral	Control	38.5±0.10	38.7±0.15	38.7±0.09	38.9±0.11	38.7±0.12	Day***, T.Day***
		Herb	38.7±0.10	38.8±0.15	38.7±0.09	38.9±0.11	38.5±0.12	
	Heat Stress	Control	38.9±0.10	38.6±0.15	38.4±0.09	39.1±0.11	39.1±0.12	
		Herb	38.8±0.10	38.7±0.15	38.4±0.09	39.0±0.11	38.6±0.12	
PM	Thermoneutral	Control	39.5±0.12	39.0±0.08	39.3±0.09	39.1±0.12	39.0±0.09	T***, Diet*, Day***, T.Day*, T.Diet.Day**
		Herb	39.2±0.12	39.1±0.08	39.1±0.09	39.1±0.12	39.2±0.09	
	Heat Stress	Control	39.5±0.12	39.5±0.08	39.7±0.09	39.4±0.12	39.9±0.09	
		Herb	39.4±0.12	39.2±0.08	39.7±0.09	39.4±0.12	39.3±0.09	

*** Significant at $P<0.001$; ** Significant at $P<0.01$; * Significant at $P<0.05$; NS-Non Significant

In addition, temperature x diet ($P<0.05$), temperature x day ($P<0.001$), diet x day ($P<0.001$) and temperature x diet x day ($P<0.01$) also influenced the pulse rate, such that the difference in pulse rate between the thermoneutral and heat stress conditions was greater on day 60 than other days. In contrast, the herbal diet reduced the pulse rate in all those days of recording other than day 0 (Table 6).

Rectal temperature (RT)

Rectal temperature was lower in the morning than in the afternoon (38.7 vs. 39.3 °C, $P<0.001$). Rectal temperature in the morning was different between days, being lowest on day 30 and highest on day 45 (38.7, 38.7, 38.6, 38.9 and 38.7 °C for days 0, 15, 30, 45, and 60, respectively, $P<0.001$) (Table 7). While there was no significant effect of temperature on rectal temperature in the morning (38.7 vs. 38.8 °C for thermoneutral and heat stress conditions, respectively, $P=0.35$) there was a significant interaction between temperature and day of treatment such that the rectal temperature was higher in sheep under thermoneutral conditions on day 30 (38.7 vs. 38.4 °C) while the converse occurred on day 60 (38.6 vs. 38.9 °C). The rectal temperature in the afternoon was increased during heat stress (39.2 vs. 39.5 °C for thermoneutral and heat stress conditions, $P<0.001$) and was reduced by the herbal supplement (39.4 vs. 39.3 °C for the control and herbal diet, respectively, $P=0.03$) (Table 7). Rectal temperature in the afternoon was

different between days, being lowest on day 15 and highest on day 30 (39.4, 39.2, 39.4, 39.2 and 39.3 °C for days 0, 15, 30, 45, and 60, respectively, $P=0.002$). However, there were significant temperature x day ($P=0.017$) and temperature x diet x day ($P=0.003$) interactions such that the difference in rectal temperature between the thermoneutral and heat stress conditions was greater on day 30 and day 60 than on day 0, especially in those consuming the herbal diet on day 60 (Table 7).

Effects of heat stress and herbal powder supplementation on different heat shock protein expression pattern in Kenguri sheep

Expression pattern of HSF1

The effect of heat stress and herbal powder supplementation on *HSF1* expression pattern in Kenguri sheep was described in Fig. 4. Among the different groups, the significantly ($P<0.01$) higher expression of *HSF1* was recorded in the KHS group, followed by KHSS, KCS and KC groups, respectively. Expression of *HSF1* was increased by heat stress (1.01 vs. 4.11; $P<0.01$) but this significant increase was reduced by supplement with herbs (4.11 vs. 1.93; $P<0.01$) (Fig. 4). Further, there was a significant ($P<0.01$) interaction between heat treatment and diet such that herbal supplementation reduced *HSF1* expression during heat

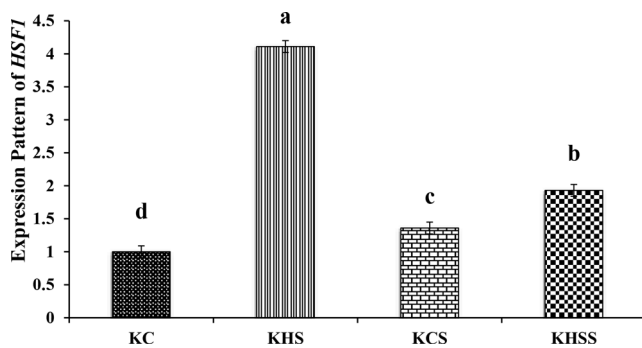


Fig. 4 Effect of heat stress and herbal product supplementation on *HSF-1* mRNA expression pattern in Kenguri sheep. KC (Kenguri Control), KHS (Kenguri Heat Stress), KCS (Kenguri Control Supplement) and KHSS (Kenguri Heat Stress Supplement)

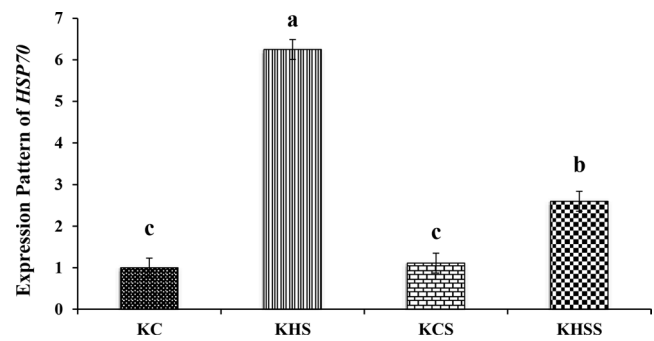


Fig. 6 Effect of heat stress and herbal product supplementation on *HSP70* mRNA expression pattern in Kenguri sheep. KC (Kenguri Control), KHS (Kenguri Heat Stress), KCS (Kenguri Control Supplement) and KHSS (Kenguri Heat Stress Supplement)

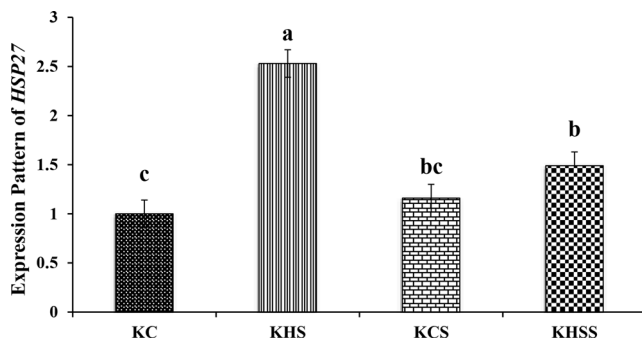


Fig. 5 Effect of heat stress and herbal product supplementation on *HSP27* mRNA expression pattern in Kenguri sheep. KC (Kenguri Control), KHS (Kenguri Heat Stress), KCS (Kenguri Control Supplement) and KHSS (Kenguri Heat Stress Supplement)

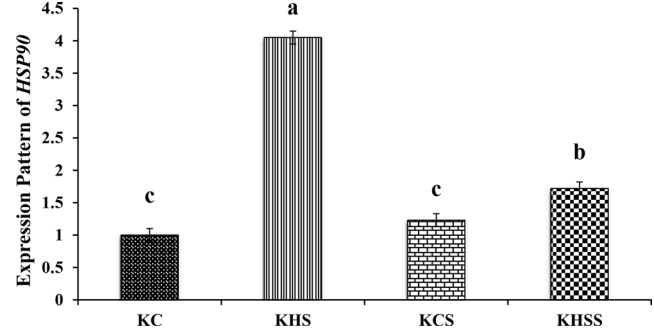


Fig. 7 Effect of heat stress and herbal product supplementation on *HSP90* mRNA expression pattern in Kenguri sheep. KC (Kenguri Control), KHS (Kenguri Heat Stress), KCS (Kenguri Control Supplement) and KHSS (Kenguri Heat Stress Supplement)

stress (4.11 vs. 1.93) but not under thermoneutral conditions (1.01 vs. 1.36).

Expression pattern of HSP27

The effect of heat stress and herbal powder supplementation on *HSP27* expression pattern in Kenguri sheep was described in Fig. 5. Among the different groups, the significantly ($P < 0.01$) higher expression of *HSP27* was recorded in the KHS group, followed by KHSS, KCS and KC groups, respectively. Expression of *HSP27* was increased by heat stress (1.00 vs. 2.53; $P < 0.01$) but this significant increase was reduced by supplement with herbs (2.53 vs. 1.49; $P < 0.01$) (Fig. 5). Further, there was a significant ($P < 0.01$) interaction between heat treatment and diet such that herbal supplementation reduced *HSP27* expression during heat stress (2.53 vs. 1.49) but not under thermoneutral conditions (1.00 vs. 1.16).

Expression pattern of HSP70

The effect of heat stress and herbal powder supplementation on *HSP70* expression pattern in Kenguri sheep was described in Fig. 6. Among the different groups, the significantly ($P < 0.01$) higher expression of *HSP70* was recorded in the KHS group, followed by KHSS, KCS and KC groups, respectively. Expression of *HSP70* was increased by heat stress (1.00 vs. 6.25; $P < 0.01$) but this significant increase was reduced by supplement with herbs (6.25 vs. 2.60; $P < 0.01$) (Fig. 6). Further, there was a significant ($P < 0.01$) interaction between heat treatment and diet such that herbal supplementation reduced *HSP70* expression during heat stress (6.25 vs. 2.60) but not under thermoneutral conditions (1.00 vs. 1.11).

Expression pattern of HSP90

The effect of heat stress and herbal powder supplementation on *HSP90* expression pattern in Kenguri sheep was described in Fig. 7. Among the different groups, the significantly ($P < 0.01$) higher expression of *HSP90* was recorded in the KHS group, followed by KHSS, KCS and KC groups,

respectively. Expression of *HSP90* was increased by heat stress (1.00 vs. 4.05; $P < 0.01$) but this significant increase was reduced by supplement with herbs (4.05 vs. 1.72; $P < 0.01$) (Fig. 7). Further, there was a significant ($P < 0.01$) interaction between heat treatment and diet such that herbal supplementation reduced *HSP90* expression during heat stress (4.05 vs. 1.72) but not under thermoneutral conditions (1.00 vs. 1.24).

Expression pattern of HSP110

The effect of heat stress and herbal powder supplementation on *HSP110* expression pattern in Kenguri sheep was described in Fig. 8. Among the different groups, the significantly ($P < 0.01$) higher expression of *HSP110* was recorded in the KHS group, followed by KHSS, KCS and KC groups, respectively. Expression of *HSP110* was increased by heat stress (1.03 vs. 3.48; $P < 0.01$) but this significant increase was reduced by supplement with herbs (3.48 vs. 1.44; $P < 0.01$) (Fig. 8). Further, there was a significant ($P < 0.01$) interaction between heat treatment and diet such that herbal supplementation reduced *HSP110* expression during heat stress (3.48 vs. 1.44) but not under thermoneutral conditions (1.03 vs. 1.29).

Discussion

The results obtained from the study demonstrated the inherent potential of Kenguri sheep to adapt to high levels of heat exposure through the alteration of both their physiological and molecular mechanisms.

The temperature-humidity index (THI) value recorded clearly projected the extremely severe heat stress being experienced by the KHS and KHSS groups of sheep. Any value over 78 as per the McDowell (1972) formula was considered extremely stressful. The maximum estimated THI value in the current study recorded was 93.3, and such a value would have ensured that both KHS and KHSS groups were exposed to extremely severe heat stress. A similar study conducted in goats in the same climatic chamber recorded a similar high range of THI was observed in the heating chamber (Devapriya et al. 2021; Spandan et al. 2022). Thus, the microclimate in the heating chamber proved that the heating chamber housed experimental animals experienced extreme heat stress. The ability of the Kenguri sheep to tolerate such conditions indicates their climate resilience. Nevertheless, there were pronounced physiological and biochemical responses to heat stress, especially in the sheep that didn't receive the herbal supplement. Further, the average THI during the study period at 14.00 h in a thermo-neutral chamber was 68.9. As per the McDowell (1972) formula

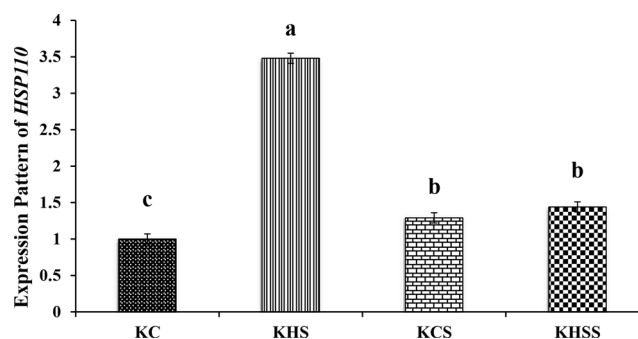


Fig. 8 Effect of heat stress and herbal product supplementation on *HSP110* mRNA expression pattern in Kenguri sheep. KC (Kenguri Control), KHS (Kenguri Heat Stress), KCS (Kenguri Control Supplement) and KHSS (Kenguri Heat Stress Supplement)

any value below 72 was considered to be a comfort zone. Thus, both the KCS and KCSS animals were proved to be under comfortable conditions in the TNZ chamber during the entire study period. Similar such reports as described above has also recorded comfort zone in the TNZ chamber (Devapriya et al. 2021; Spandan et al. 2022).

The physiological mechanisms are among the primary mechanisms employed by the heat stressed small ruminants to dissipate heat when subjected to extremely severe heat stress conditions (Marai et al. 2007; Sejian et al. 2010; Joy et al. 2020). Further, the herbal powder supplementation significantly reduced RRA recorded in KHSS group as compared to KHS group reflecting the beneficial effects of herbal powder supplementation. The RR is considered an important biomarker for heat stress and the reduced level of RR in herbal supplemented group during heat stress exposure could serve as a reliable feeding technology for reducing heat stress complications in Kenguri sheep. Similar to our finding, Hashemzadeh et al. (2023) also reported a beneficial effect of a phytogenic-rich herbal mixture in relieving heat stress in rams. In addition, Liu et al. (2016) also reported similar beneficial effects of herbal supplements in heat-stressed lambs. Likewise, Joshi et al. (2023) also proved herbal supplements to be an effective heat alleviation strategy in lamb raised in heat stress condition. Similarly, herbal supplements reduced physiological parameters in heat stressed ewes reared in sub-tropical regions (Al Mufarji et al. 2022). There is increasing evidence that plant-derived phytochemicals can improve antioxidant status and protect against the oxidative stress that occurs during heat stress. For example, Shakeri et al. (2020) found that a phytochemical rich sugar cane molasses extract improved growth and reduced the physiological responses to heat stress in chickens. The mixture of herbs chosen for the present study are all relatively high in phytochemicals. It is likely that plant bioactives may benefit host animals through their immunomodulatory, anti-inflammatory, antimicrobial, and

antioxidant effects. Some plant bioactive may also improve the integrity and functionality of the gastrointestinal tract in animals challenged with stressors (Patra et al. 2019).

In a similar study conducted using Merino ewes, DiGiacomo et al. (2016) and Dunshea et al. (2017) observed the beneficial role of betaine supplements in reducing respiration rate in heat-stressed animals. From the above findings, it is evident that respiration rate could serve as a reliable biomarker for quantifying heat stress response as well as, it could serve as a reliable indicator for validating heat stress-relieving effects of herbal supplements in Kenguri sheep.

Similar to the findings regarding respiration rate, heat stress significantly increased pulse rate, which was substantially reduced by herbal supplements. This was evident from the significantly increased PRA in the KHS group as compared to the significantly lower level in the KHSS group. Similar to our findings, Sejian et al. (2010) and Singh et al. (2016) have reported significantly higher pulse rates in heat-stressed different indigenous breeds of sheep. The herbal supplements reduced PRA in KHSS animals suggest that, like RRA, PRA could also serve as a reliable indicator for understanding the beneficial effects of such supplements.

The RTA also showed a similar trend to that of RRA and PRA in the current study. Further, this was similar to the study findings reported in various breeds of sheep kept under heat stress exposure (Rashid et al. 2013; de Sousa Silva et al. 2024). The increased rectal temperature in heat-stressed animals suggests the highly active physiological mechanisms for heat dissipation (Marai et al. 2007). Similarly, Srikandakumar et al. 2003 suggested rectal temperature to be a good index of deep body temperature reflecting the thermal balance. They could play a role in evaluating the impact of heat stress on sheep. Further, the study also proved the effectiveness of herbal supplements in reducing the heat stress-induced rectal temperature in Kenguri sheep. This shows the beneficial effect of the herbal supplements used in the study for heat stress alleviation in Kenguri sheep.

The molecular mechanism governing adaptation is one of the crucial strategies exhibited by heat-stressed animals (Sejian et al. 2019). The heat stress significantly increased the expression pattern of *HSF-1*. Similar to our finding, Samara et al. (2024) reported a pronounced up-regulation of *HSF-1* in Najdi lambs. The *HSF-1* was established to be the primary regulator of heat shock response, and exposure to heat stress activates this heat shock transcription factor to stimulate HSP production (Das et al. 2016). Further, Sejian et al. (2019) also noted *HSF-1* to be one of the biomarkers for assessing heat resilience in sheep. The significantly lower level of *HSF-1* in the KHSS group as compared to KHS could be attributed to the beneficial effect of herbal supplements used in the study for lowering the heat stress-induced up-regulation of *HSF-1*.

Heat stress significantly increased the expression pattern of *HSP27* in the current study. Thus, *HSP27* could be a reliable marker for assessing heat stress in indigenous sheep. Similarly, Mohanarao et al. (2014) also suggested that *HSP27* could be a useful indicator for assessing in vitro stress response in goats. Likewise, Kapila et al. (2013) also reported a significant increase in the expression of *HSP27* in buffalo mammary epithelial cells during heat stress exposure. Further, herbal supplementation reduced the heat stress-induced upregulation of *HSP27*, indicating a beneficial role of herbal supplementation during heat stress exposure in Kenguri ewes.

The heat stress-induced significant upregulation of *HSP70* in the study was similar to the findings reported in Pelibuey sheep by Romero et al. (2013), Balkhi sheep by Khan et al. (2020) and Munjal sheep by Kumar et al. (2025). Further, the *HSP70* gene has been established as the ideal molecular marker for quantifying heat stress response in other livestock species (Aleena et al. 2018). In addition, the herbal supplementation in the study significantly reduced the heat stress-induced increase in the expression pattern of *HSP70*, establishing the potential application of these herbs in relieving heat stress in Kenguri sheep. The significantly higher and lower expression of *HSP70* in KHSS as compared to KCS and KHS groups, respectively, establishes the heat stress protective effects of the herbal supplements used in this study.

A similar trend to that of *HSP70* was established for *HSP90* also. Likewise, Lu et al. (2019) also reported an increase in the mRNA expression of *HSP70* and *HSP90* in heat-stressed Hu sheep. This shows that both *HSP70* and *HSP90* can act as molecular markers for quantification of heat stress in Kenguri sheep. This claim was strengthened by the findings of Sejian et al. (2019), who reported that both *HSP70* and *HSP90* can act as biomarkers to assess the resilience capacity of sheep and goats to heat stress. Further similar to *HSP70* expression, the heat stress-protective effects of herbal supplements also was established in the KHSS group, with significantly lower *HSP90* expression recorded in this group as compared to the KHS group. The expression pattern of *HSP110* also showed the same trend of increased expression in heat-stress Kenguri sheep. This finding was similar to the findings of Lu et al. (2020) and Zhang et al. (2024) in Hu sheep. These authors also reported significantly higher expression of *HSP110* in Hu sheep after heat stress exposure. This shows that *HSP110* could also serve as a reliable indicator of heat stress in Kenguri sheep. In addition, the herbal supplementation also significantly reduced the *HSP110* expression. This was evident from the significantly lower expression of *HSP110* in the KHSS group as compared to the KHS group ewes. This reflects the heat stress-protective effects of herbal supplements in the

study by reducing the expression pattern of *HSP110* which was considered one of the important heat stress biomarker in sheep.

Conclusion

Based on the study results, it could be inferred that the Kenguri breed activated both their physiological and molecular mechanisms to adapt to heat stress challenges. Based on the results, the study established the RR, PR, RT, *HSF1*, *HSP27*, *HSP70*, *HSP90* and *HSP110* as biomarkers for quantifying heat stress response in Kenguri sheep. Further, supplementation of herbal powder significantly reduced the adverse impact of heat stress on the physiological and genetic traits studied. While this study points towards the heat stress protective effects of the herbal supplement, future research should take an effort to identify the active constituents of the herbal supplement in order to elucidate key mechanisms by which they mitigate the adverse heat stress effects in Kenguri ewes. Further, the study points towards using this herbal powder supplement in other sheep breeds with an intention to test their potential in general for relieving heat stress in sheep. Thus, the positive effect of the herbal supplement on the variables studied clearly established their potential for improving climate resilience in Kenguri ewes, thereby contributing to sustainable sheep production in the context of climate change.

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Author contributions VS, CD, FRD- Conceptualization; EBR & CD- Conducted the experiment, analysis of variables; VS, MVS, JN, FRD & AS – Data analysis; EBR, CD, SVP and MVS – Drafted the manuscript; VS, JN, AS & FRD – reviewed and edited the manuscript for submission.

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Data availability The data presented in this study are available on request from the corresponding author.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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